

MassMarsh
Standard Operating Procedure:
Transect-Based Collection of Vegetation Data

1. Scope and Application

A standard set of procedures is established to set up a monitoring effort to assess salt marsh vegetation using permanent plots along transects to characterize the plant community for assessment of the marsh conditions or pre and post restoration actions. The data produced can be statistically reduced to yield an overall picture of wetland vegetation and changes over time for interpretation regarding marsh health or vulnerability as part of a comprehensive wetlands assessment (e.g., Moore 2017, Kutcher 2024, NERRS 2024). Alternatively, these data can be divided into geographic or management units for a particular marsh to evaluate restoration actions (e.g., restoration; Neckles and Dionne 2000, Smith and Warren 2012, Vincent et al. 2014) supporting a monitoring and adaptive management program. In addition, vegetation data produced using this protocol can be used to develop remotely sensed metrics for management (Jackson et al. 2025) and conversely, remotely sensed data collections can improve documentation and interpretation of vegetation changes.

2. Summary of Approach

Vegetation reflects the hydrology and soil development in salt marshes and is an important indicator of marsh function (habitat, biodiversity, coastal protection). By following changes in the plant community, we can assess the overall condition of the marsh and whether restoration actions are supporting plant development indicative of healthy marshes. The species, density and vigor of vegetation in salt marshes are broadly used to characterize marsh health; predicting whether marsh areas will be sustainable and resilient or vulnerable and at-risk from such threats as sea levels rise. Vegetation assessments over time yielded the first evidence of vegetation change in response to sea level rise (Warren and Niering 1993, Donnelly and Bertness 2001). Vegetation data can be especially valuable when they are used in conjunction with tidal range and marsh surface elevation to establish the tidal frame that encompasses existing elevations of vegetated marsh. Nevertheless, elevation does not always determine flooding regime across large expansive marshes since hydrological impairments can lead to severe waterlogging at higher elevations and well-drained conditions at relatively lower elevations.

The transect approach has been shown to be valuable in many studies and is based on a series of monitoring guides and protocols (Neckles and Dionne 2000, Roman et al. 2001, Neckles et al. 2002, Moore 2017, Rocks and Stevens 2020, NERRS 2024). Transect locations may be selected randomly for general, long-term evaluations of a marsh or directed to targeted, non-random

locations when focused on a specific restoration action. The choice of transect number and length is sensitive to the amount of effort and resources available to characterize vegetation conditions. In addition, ease of access, efficiency of data collection, and adequate representation of all visually distinctive vegetation areas are all important considerations when selecting transect locations. Some flexibility will be needed for setting up permanent monitoring plots in different types of marsh with varying geographies to achieve monitoring goals (a non-biased assessment of condition for status and trends, threats, or restoration performance).

3. Materials and Equipment

Before leaving for the field the Field Crew Leader will confirm the following equipment is available, charged and in operating condition:

- A. Real-Time Kinematic GPS and related accessories as described in Appendix A-6C.
- B. Extra batteries for antennae and GPS
- C. Smart Phone with camera and Google Earth
- D. Waterproof Datasheets and clipboard, including field notebook
- E. PVC or wooden stakes to mark transect endpoints
- F. Reel tape (50 or 100m)
- G. Quadrats and meterstick ($0.71 \times 0.71\text{m} = 0.5\text{m}^2$ and $0.25 \times 1\text{m} = 0.25\text{m}^2$)
- H. Erasable white board and markers
- I. Appropriate clothing, food, and water during field work (see safety list)

4. Calibration & Training

All field crew leaders should have experience conducting research in salt marshes and should be familiar with plant species needed to classify salt marsh plots. Field technicians should be trained and supervised by crew leaders. Monitoring teams should have access to field guides (Tiner 1987, updated in 2009). Some simple field guides that are not comprehensive are available on-line: [Life-in-New-Hampshire-Salt-Marshes—A-Quick-Reference-Field-Guide](#). All equipment will be calibrated and maintained according to manufacturer recommendations.

Procedures

5. Sampling Designs

The goals of the Mass Salt Marsh Monitoring project include both performance and ecosystem-based monitoring; therefore the designs and activities have differences (Table 1). Specifically,

transects for ecosystem monitoring and monitoring to assess mitigation of tidal restrictions are set up with a random stratified approach whereas the performance-based monitoring transects are typically located where the greatest changes are expected from implementation of restoration actions on the marsh platform. The ecosystem-based monitoring design characterizes the vegetation over time from the seaward to landward edges and considers sampling beyond the landward edge (the marsh upland ecotone) to capture the process of marsh migration driven by sea level rise. It is expected that these sites are sampled over a long-term time span, possibly 50 years. To assess changes over time, it is important to assign zones to the marsh based on recognizable plant communities (Burdick et al. 2020) so that plots falling into the original zones can be combined for statistical analysis. Transition of vegetation as one zone becomes another can also be identified as an ecotone and these data can be analyzed separately.

In contrast, performance monitoring will typically use a Before/After Control/Impact (BACI) design that is spatially focused on restoration actions to show benefits or issues with restoration over relatively short periods (5 or 10 years). Assigning marsh zones as well as defining transect plots expected to be within the zone of influence from restoration actions are important when the data are analyzed.

Table 1. Similarities and differences between ecosystem-based and performance-based monitoring designs using vegetation transects.

	Ecosystem-Based	Performance-Based
Community focus	Zones and Ecotones	Zones and Impact areas
Monitoring objective	Characterize community, identify emerging threats and vulnerabilities	Show change from actions
Results over time	Change in area or resilience	Benefits and problems
Statistical Framework	Regression over time	BACI: Before-After-Control-Impact
Statistical methods	Charts, Univariate, Linear Mixed Models, ANOSIM/SIMPER	ANCOVA, time*action interaction
Design	Random stratified transects lower to upper marsh edge plus potential migration space	Transects within restoration areas (runnel and ditch) as replicates; Transects across sediment placement or culvert removal as replicates; Transects across reference and no-action control areas
Timing	No set date; annual or less frequent	Pre-restoration (Year -1);

		Post-Restoration (Years 0–5); Annual or less frequent
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6.1 Establishment of Transects and Plots for Ecosystem-based Monitoring:

Monitoring to determine marsh condition: Following selection of a marsh for monitoring, aerial imagery can be used to outline the specific area of interest and determine the locations of transects relatively quickly and without bias. In general, it is desirable to characterize the vegetation of the entire marsh, from seaward edge to upland ecotone (and potentially into the marsh migration space) using a set of transects that run from the seaward edge or major tidal creek into the upland. Transect starting locations should be randomly selected by obtaining a set of random numbers that represent the entire creek length or seaward length of the marsh. Be sure to avoid patterns and transects closer than 10 m. To ensure fair characterization of the entire marsh area to be sampled, it is suggested the marsh be broken up into 2 to 3 segments and then the locations of 1-3 transects be randomly selected for each segment (Figure 1). If low salinity marsh areas appear to be important, they should be recognized as separate zones and may also be included. The guidance for the number of plots is area-independent (James Pirri et al. 2007); the scale of sampling effort in marshes over 100 acres in area may be increased, with a focus on additional strata for very large systems (see Figure 1).

Steps to select transect and plot locations stratified along a salinity gradient:

- 1) Assign a scale for the midline of the main creek or along seaward edge (if possible)
- 2) For each segment, select 10 random numbers out of the entire length (e.g., Figure 1 mid-marsh segment = 433M: 153,40,291,387,216,386,391,412,315,25)
- 3) Eliminate points that are too close to other points. (e.g., mid-marsh segment; keep 40,153,291,387,216,412,315,25; but 386,391 are eliminated)
- 4) Select the first three and arrange in ascending order (40, 153, 291) and alternate between sides (unless one side of the marsh is inaccessible)
- 5) Continue for the remaining two segments
- 6) Make adjustments to transects and plots to ensure a reasonable number of samples for each marsh zone or feature that is a focus of the monitoring (typically low marsh, high marsh and upper edge). For example, transect end at 291 m was switched to begin on the southern edge so that more low marsh plots would be included.
- 7) If some of the transect locations are found to be unacceptable in the field, have the next two random number locations for each segment ready so they can be set up in the field.
- 8) On the map, draw a line from each transect starting point perpendicular to the upland on the image and estimate the number of plots, 10 m apart, that will fit along the transect.

Typically transects are 40 to 100 m in length and should, if possible, begin with a low marsh plot and end at the upper marsh edge (Figure 2).

- 9) The location of the first plot on the seaward or creek edge should also be determined by a random number that will put the plot within the lowest plant community (low marsh) at a random distance from the seaward edge (Moore 2017). The seaward end of the transect may be marked with a pvc post in the high marsh away from the dynamic lower marsh where posts may be frequently lost.
- 10) The distance between plots along a transect should be 10m or greater to avoid autocorrelation (Elzinga et al. 2001). This rule of thumb can be relaxed when crossing marsh zones (low to high, high to upper edge). If working in very small marshes (e.g., southeastern MA), plots may need to be closer than 10m, but the autocorrelation issue should be acknowledged. Quadrat size for grass-dominated systems can be either 1.0 m² or 0.5m². The number of plots of the community of interest (e.g., high marsh) should sum to roughly 20 plots (10 m² sampling area; for more guidance on number of plots see Appendix). If not, add another transect until 20 plots are reached. In the example, Figure 2 shows this arrangement yields 20 plots in the high marsh so subtle changes can be detected. In addition, there will be 9 low marsh and 7 upper edge plots which appears to be sufficient to capture changes over the long term. If a more balanced design is sought, select more transects and increase the space between plots (e.g., 20 m).

Plot locations may fall on embankments, forb pannes, ditches, creeks, pools or bedrock outcrops. Although one might exclude these odd plot types from statistical analysis, changes in these plots may be illuminating (e.g., a pool that drains, a creek that shifts laterally). In the high marsh, monitoring programs should identify high-diversity forb pannes, in contrast to pannes indicative of marsh deterioration (mostly short-form *S. alterniflora*) and pools (unvegetated). Submerged vegetation can thrive in pools (e.g., *Ruppia maritima*) and increase habitat value and so should be assessed for cover.

Stretching a reel tape between transect ends allows each plot to be relocated to within 10-30 cm without the need for marking each plot. The reel tape should be held close to the ground on windy days to keep a straight transect and avoid permanently stretching the tape. The exact position of transect ends should be determined by RTK GPS; in case a post is lost it can be reset. The elevation of a nearby temporary benchmark can be determined at the same time to provide NADV 88 elevation so that the absolute elevations of the plots can be determined (see Sediment and Elevation SOP).

6.2 Monitoring to determine effectiveness of marsh restoration:

Due to the small size of many of the features undergoing restoration (ditch width, runnel) a smaller quadrat size (e.g., 0.5m²) is justified. However, more samples will be required to

characterize the vegetation, and if differences over space or time actually occur, the ability to find differences (i.e., power) will be reduced.

Restoration of tidal exchange: If the restoration plan includes restoring tidal exchange to an area, establish transect locations that should reflect expected changes in the vegetation over time (Figure 1). Please include areas currently uplands but expected to become tidally influenced following tidal restoration using hydrologic model outputs of the new expected tidal range with LiDAR maps or SLAMM maps (with barriers removed) for a chosen future date (e.g., 2070).

Restoration of tidal exchange through culvert or bridge replacement can be examined using the previous procedure for transect selection, being sure to include upper marsh areas even if transects are narrow (the 10 m ‘rule’ can be ignored for the distance between high and upper marsh plots and between upper and upland plots because a different plant zone or community is being sampled). Upper marsh plots should have some marsh obligate plants but also some upland or marsh edge plants (e.g., *Panicum virgatum*). See Glossary for definition of zones

Sediment deposition to increase elevation: If the restoration plan includes thin layer sediment deposition, establish transects that extend from drainage paths or marsh seaward edges across to the upland edge. Upland edges may not be an important component if well-removed from the sediment deposition area (although higher elevation plots might reveal plant compositions and species that should be useful if the elevation of the deposition area has high spots). Extra plots should be added on the edges of drainage features because often these areas repopulate better than those more distant (Wasson et al. 2024).

Restoration of surface hydrology using ditch remediation, drainage enhancement and habitat islands: Ditch remediation transects should run perpendicular (normal) to the ditches and include remediated and unremediated ditches. The sample replicate for ditch remediation is the ditch, so you might want several distinct ditches that will be remediated that cross the transects selected, as shown in Figure 3. Unremediated ditches are considered controls and should equal the number of remediated ditches for a balanced design. Alternatively, another area of marsh with no remediated ditches might be selected as a control, since the effects of the ditches may extend several meters. Since only four control ditches are shown in Figure 3, a balanced design would require another pair of transects established nearby to capture conditions across four more unremediated ditches.

The assessment focus is on the ditches and bringing the water table up between ditches. Therefore, plots within the ditch center and on the marsh platform (roughly equidistant between ditches) will provide information about plant colonization of the ditch center as well as plant responses to rising water tables. If possible, begin and end each transect with a high marsh plot between two unremediated ditches so the next platform will lie between an

unremediated and a remediated ditch (Figure 3). This set-up will provide several unrestored platform plots to serve as controls. Because ditches can be as narrow as 25 cm, paired 0.25 x 1.0m plots are used on either side of the transect (Figure 3 inset B, Figure 4). Continue to use 0.5 m² quadrats on the platform, since platform and ditch center plots are analyzed separately. The quadrat size chosen must be adhered to in future monitoring years.

Areas of the marsh that are to receive treatments to improve drainage (runnels, ditch clean-out) should run from the seaward edge to the upland marsh edge. This design allows these sites to be used as long-term monitoring areas in the future. If this is not possible, have one transect endpoint reach the upper marsh and the other reach what is considered 'healthy marsh' (high marsh graminoids dominating). These healthy marsh plots (and in some cases the plot prior to the upper end of the transect) can provide a good target reference for the pool and panne areas that will be drained. For runnels draining relatively deep pools, single or paired plots along the edges of the pools can be added to the sampling effort (Figure 3, inset A and Figure 4). Tracking vegetation change across deep pools may also benefit from remote sensing. To assess how the site with enhanced drainage responds, no-action control transects should be established at locations that will not be affected by the restoration. The healthy marsh target plots and the no-action control plots provide bookends for the treatment plot metrics (e.g., % cover of high marsh graminoids).

Habitat islands are not assessed with transects but use a combination of measures, including island size, vegetation and elevation. Island area is measured with a reel tape across the widest (length) and narrowest (width) dimensions and calculated using the ellipse (area = length * width * π). Four vegetation plots (0.5 m²) that assess percentage cover by species can be collected from each island. Elevations are measured at the top of the upper slope break at four locations around the island (Figure 5) and tied to temporary benchmarks (see Sediment and Elevation SOP).

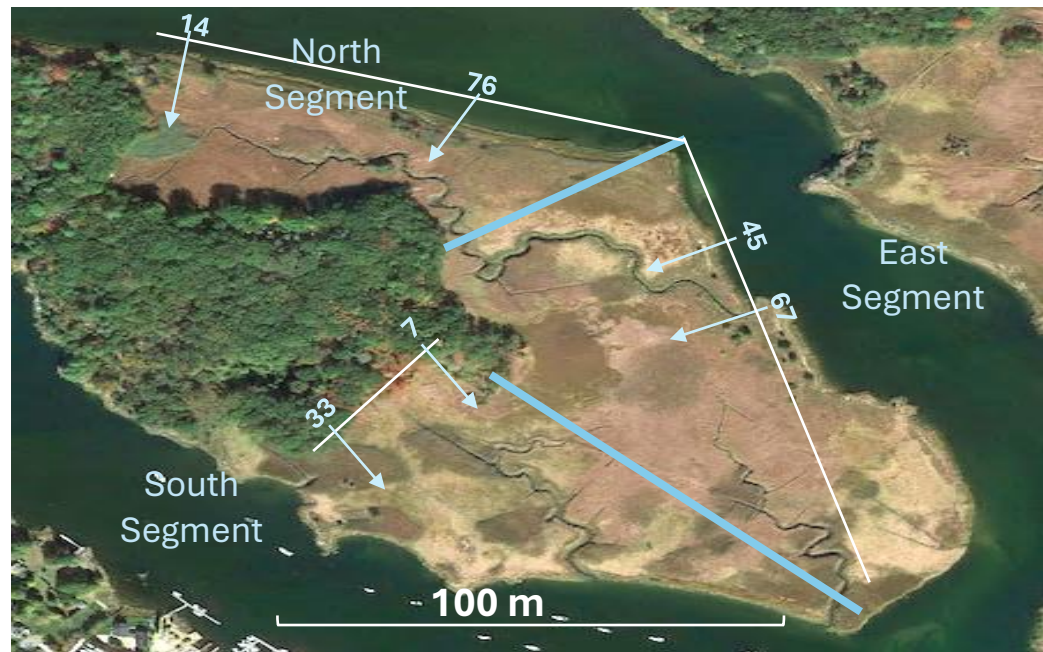


Figure 1. Examples for long-term evaluation of marsh condition using a stratified random design to select transect locations in A) A riverine marsh of approximately 95 acres; and B) An embayment marsh surrounding a headland.

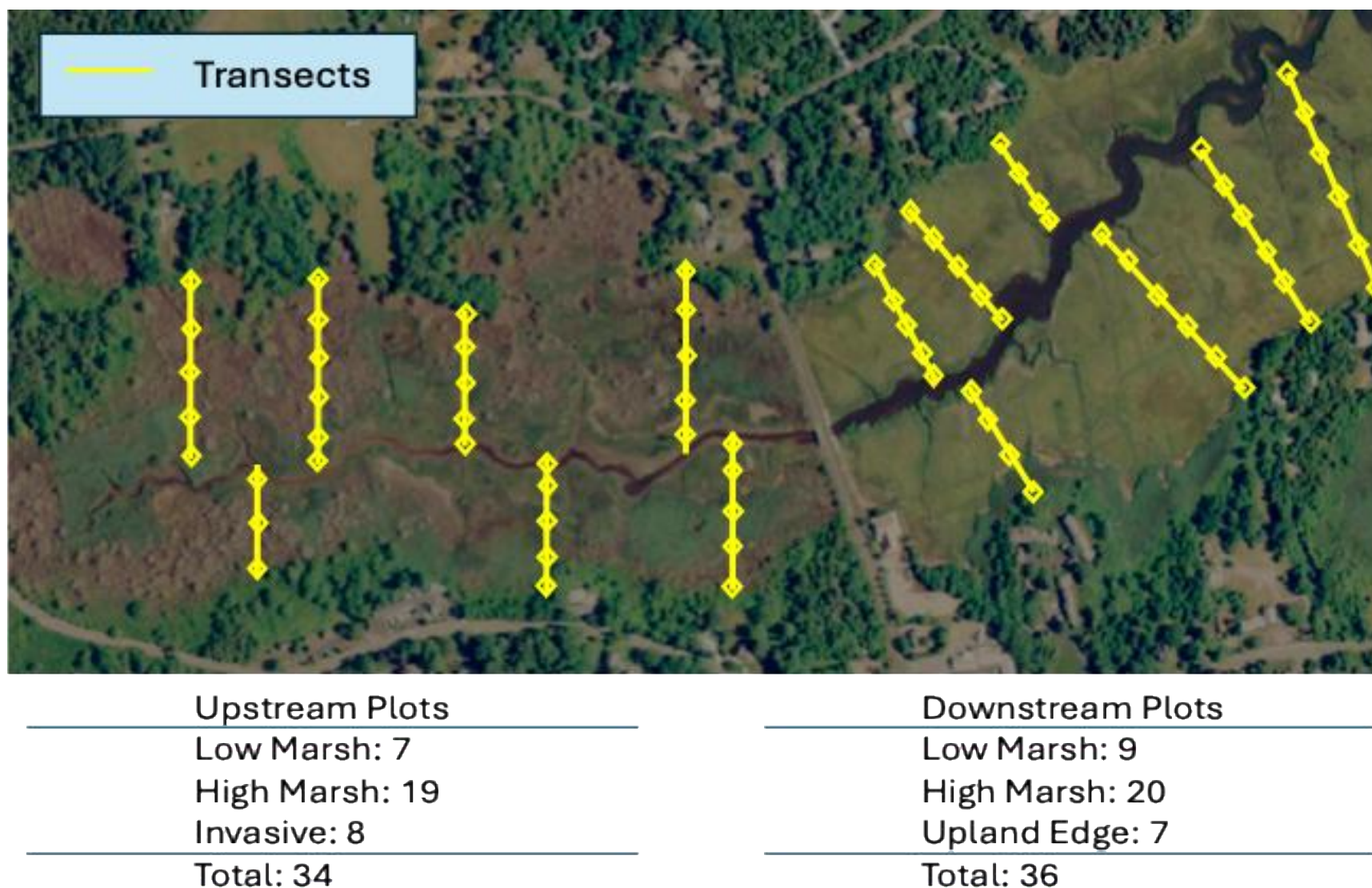
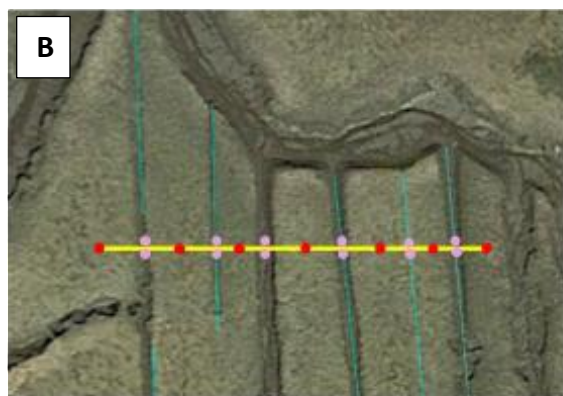


Figure 2. Example of a set of transects established to document salt marsh vegetation response to mitigation of restricted tidal exchange.



Optional plots along pool edges



Plots on platform (red) and ditch centers (pink) shown



Four Ditch Remediation Transects
 Total of 46 plots
 Ditch remediation replicates: 8
 Marsh platform: 25 plots
 Remediated ditch centers: 14 plots
 Unremediated ditch centers: 7 plots

Five Drainage Enhancement Transects
 Total of 27 plots, 10-meter intervals
 Drainage enhancement reps: 5
 Reference high marsh: 5 plots
 Pannes: 20 plots
 Pools: 2 plots

Figure 3. Transect arrangement for a portion of marsh being restored with single channel hydrology (parallel ditches remediated to reduce soil over-oxidation, runnels added to create drainage to reduce soil waterlogging). Plots on marsh platform shown as red dots; ditch center paired plots are pink. Inset A shows optional pool edge plots and inset B shows ditch center and platform plots.



Figure 4. Left: One of two paired plots (0.25m^2 in size) in the ditch centers, showing no regrowth of *S. alterniflora* within the plot. Right: Pool edge with approximate placement of optional paired plots (0.05m^2) and single plot (1m^2).



Figure 5. Habitat island or marsh mound with blue outline and orange points showing crest of slope break where elevation should be measured. These could also be the locations of the vegetation plot centers.

6.3 Sampling Approach

Vegetation assessment is typically conducted once plants have reached their maximum standing crop but before they begin to senesce. In Massachusetts, monitoring is usually conducted in a sampling window from mid-July through September for north of Cape Cod and July-August for south of Cape Cod. A few species begin to senesce before this period, notably *Juncus gerardii* (blackgrass), so inexperienced teams should begin earlier or visit sites and place several quadrats earlier.

There exist a variety of ways to assess vegetation. Fortunately, since the vegetation stature is low, we can look at a 2-dimensional plot outlined by the quadrat and estimate what we see (ocular cover). This flat plot will have cover summing to 100% and these data provide a measure of each species' ability to capture sunlight for photosynthesis. Just as important is the proportion of the plot without live plants (bare soil, rock, dead (rooted), and wrack (unrooted dead that has washed into the plot)). An objective record of each plot can be documented with a vertical downward photo of the plot (Figure 5), containing identifying data written on a white board (NERR 2024).

All the cover types and species with live vegetation are first identified in the plot without moving the vegetation. Once the monitors can identify all the salt marsh species in a plot, we can refer to images to help guide our estimates of cover (Figure 4). Many practitioners find estimation of percent cover daunting, but if the work is done in pairs, inaccuracies can be effectively reduced, paying particular attention to removing any bias.

SIDEBAR:

Depending on the statistical design of the monitoring program, several plots are often averaged to create one replicate. For example, there may be 4 plots across a transect and six transects that are considered replicates used to evaluate a certain restoration action. The transect data of the 4-plot averages should be distributed normally, thanks to the Central Limit Theorem, and have fairly small error. Alternatively, if single plots are replicates, there should be enough in the monitoring design (>10) for the central limit theorem to give confidence that the estimates of cover follow a normal distribution.

Inexperienced monitors should be given the time to become familiar with plant species and non-living cover types and work with experienced monitors for several field sessions. Monitors estimate the cover of each species: +/- 10% if species coverage is over 50%; +/-5% if 10-50%; +/-2% if under 15%; and +/- 1% if under 6%. Single shoots of very small plants receive 0.5% cover. Species hidden under taller plants are given 0.5% cover to acknowledge their presence. This approach becomes second nature after several hours of experience. Distinguishing similar species (e.g., *S. patens*, *D. spicata*, *J. gerardii*) can be challenging (see **10.0 Troubleshooting**), especially if some have senesced (e.g., *J. gerardii* or red fescue, *Festuca rubra*).

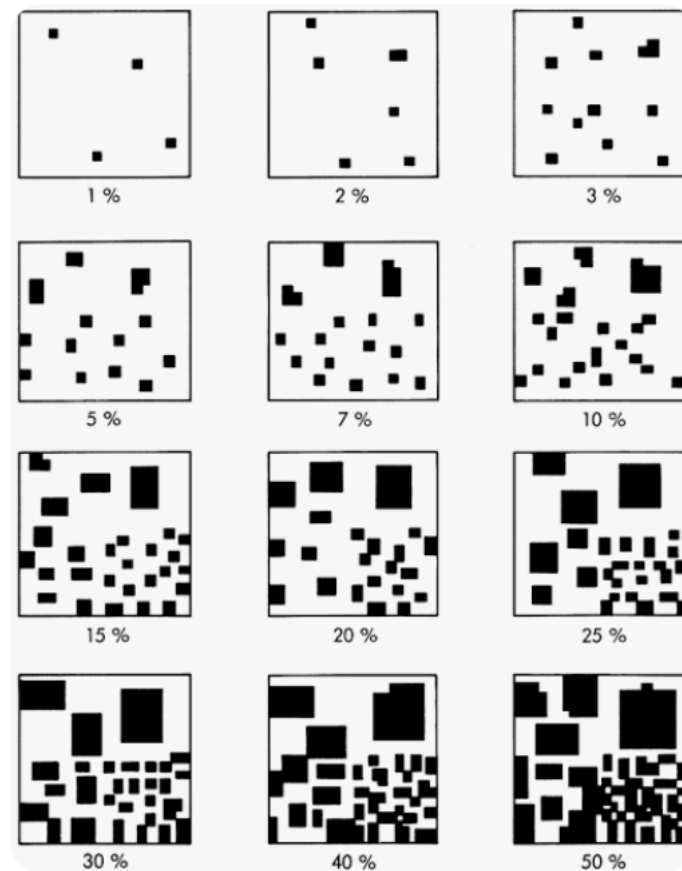


Figure 4. Representation of cover classes using four distributions from 1% to 50% cover that follow a clockwise pattern, with disaggregated cover in the lower right proceeding to a solid block in the upper right. Image from Kansas Geological Bulletin 312.

6.4 Recording data

Changes in dominance within a plot over time (e.g., short form *S. alterniflora* to *S. patens*) is a strong indicator of changes in hydrology and soil waterlogging and is assessed using percentage of cover. Cover data can also be used to generate several metrics useful for addressing changes in healthy or vulnerability or assessing restoration project performance. Additional variables can also make useful metrics (canopy height, stem density, thatch height) and collection protocols are outlined here.

1) Lay the tape and photograph the transect: Each sampling year transects are re-occupied by relocating the endpoints and stretching a reel tape between the ends. Avoid trampling vegetation on the left of the transect as you face the upland. The plot quadrat will be placed on the left of the tape so all steps should be on the right of the tape. The marsh, date, transect code and direction (e.g., E to W) is written on a white board and the transect is photographed

from each end. Photographs record general marsh condition and can lend insights to changes observed in the quantitative plant data. Picture posts can be established in limited cases, but posts will not survive many winters in much of New England. If a picture post is used, place the post in adjacent upland with a focus on the marsh area being actively managed.

2) Field data sheet and quadrat placement: A sample field data sheet is attached to this SOP (Appendix 1) and allows recording of general weather conditions and field participants as well as transect and plot information, including marsh zone. With the tape stretched along the transect, the quadrat is placed on the left side of the transect (facing the upland) with the center of the left side of the quadrat at the tape distance mark.

3) Record the cover: The two monitors, one recording, examine the vegetation, identifying the main species and estimate an overall live and non-live cover. An estimate for each cover type is said aloud and can begin with the most common or least common cover, whichever is easier. Typically, a four or five letter code for each species is used to record the data, with the percentage cover for each species and cover class present. The code is composed of the first two letters of the Genera and first three letters of the species: *Spartina alterniflora* is coded Spalt. A list of typical species and their code are provided in the Appendix. Non-living cover can include bare soil, dead attached plants (dead) unattached dead material (wrack), and rock. Algae is recorded to the lowest taxon as practicable (e.g., *Vaucheria sp.*). Algae is considered live cover and should be included in the metric 'live cover'. The proportion of the plot that is covered by water is estimated (+/- 10%), but these data are never included in the 100% cover estimate.

4) Measure plant or canopy height: Depending on specific monitoring needs, plant height can be measured (4 plants closest to each corner of the quadrat for each dominant species). Typically, plant height is measured while holding up the leaves to estimate total height to one cm. Alternatively, a simpler measure of plant height is the canopy height (also called 4/5 canopy height), estimated at the height at which 20% of the plants are taller (without holding up the plants), also measured to one cm. This simple and rapid field measure yields a comparable estimate of plant height without regard to species.

5) Measure thatch height: Because salt marsh sparrows are ground nesters and subject to flooding and nest loss, another measure of marsh vegetation quality for sparrow nesting habitat is the thatch height. Thatch height is the height above the ground of the top of the dead vegetation (if present). Birds building their nests on thatch 7 cm above the ground are much more likely to survive long enough to fledge (get ref). We know of no metric developed to use the thatch height, but the proportion of plots dominated by high marsh graminoids with at least 5 cm thatch height might be a useful metric of habitat quality for sparrows.

6) *Measure plant stem density*: Stem density of *S. alterniflora* is needed to track the colonization of remediated ditch centers. Count the number of stems >5 cm tall that lie within the 25 by 100 cm quadrat fitted into the ditch and over the center.

Species-specific stem density has not received too much attention as a metric, however it can be used with plant height and cover to estimate aboveground biomass, which could be developed into a very useful metric in the future. Three potential sizes are recommended: the full quadrat for very large plants (e.g., *Phragmites*, *Typha*), 25 by 25 cm ($1/16 \text{ m}^2$) for moderately-sized plants (e.g., *S. alterniflora*), and 10 by 10 cm (0.01 m^2) for plants with fine stems (e.g., *S. patens*, *D. spicata*, *J. gerardii*). Placement of the smaller quadrat within the larger should occur in a pre-determined location, always halfway between the large plot center and the edge of the plot adjacent to the reel tape, for easier access (Figure xx).

7) *Photograph plot*: A photograph of each plot is made shooting downward with the quadrat in place (Figure 6).

8) *Check that cover classes sum to 100%*: All cover classes are then summed by the recorder to determine if they equal 100% before removing the quadrat. This final step reduces the cases where the sum of the cover estimates does not equal 100%.

9) *Complete the transect*: Repeat steps 2-8 for each plot on the transect.

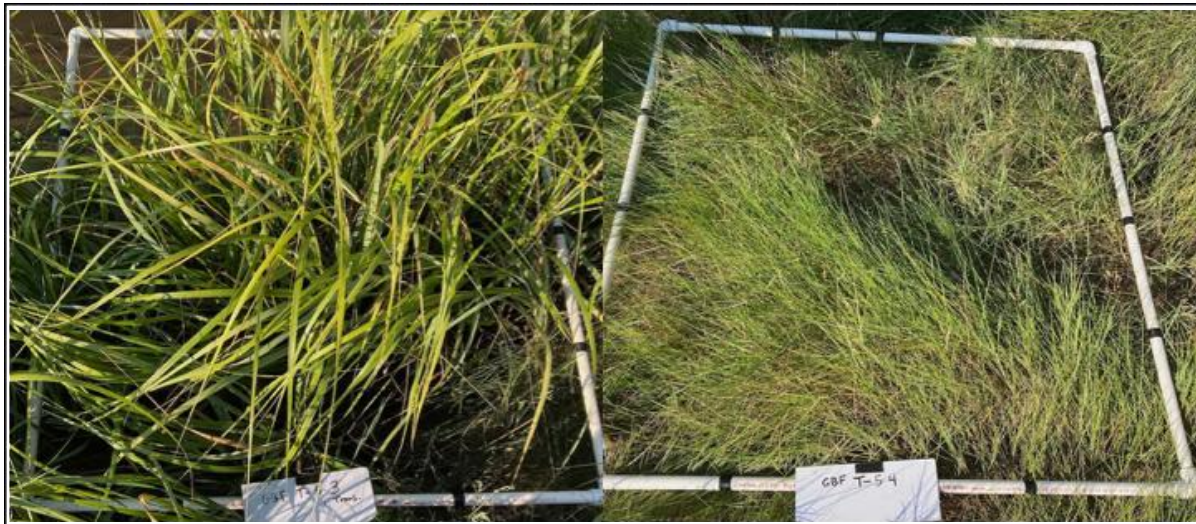


Figure 6. Photographs of 1m^2 quadrats with vegetation and whiteboard identifying site, transect and plots number as part of the NERR vegetation protocol (NERRS 2024).

6.5 Data input, forming metrics and reduction for statistical analysis

Upon return to the workplace, the field sheets should be scanned for cloud storage and the sheets stored in a file for that specific project. Data should be input to a spreadsheet for a

particular site or suite of sites with identification of the sample date, marsh, transect and plot, marsh zone and water cover followed by cover of specific types and plant species (see Appendix). Analysis can focus on specific cover types, species (e.g., Spalt), or canopy height. Useful metrics derived from the data for each plot include all living cover, non-living cover, halophyte cover, graminoid cover (Sppat+Dispi+Juger), algae cover, and plant species richness.

The spreadsheet should have a metadata tab that shows an aerial of the site with transects and any potential management actions (tidal restoration, restoration of surface hydrology, etc.).

The exact methods used must be recorded carefully in the metadata for each project, identifying site and transect naming conventions and variations in data collection protocols. Any deviations or additions to the SOP should be recorded as well as definitions of terms.

After data entry, the data should be checked using QA/QC protocols.

6.6 Protocol for Decontamination of Field Equipment

All equipment should be inspected for debris before leaving a site. Debris removed from boots and equipment should be disposed in a trash bag or on dry, high ground. When possible, equipment should be left to air dry and inspected to remove any remaining plant fragments. Waders should be sprayed with 10% bleach solution, scrubbed, and rinsed with tap water to remove any additional debris (Connors 2006).

7. Quality Assurance/Quality Control

All field crews should be led by field personnel experienced in salt marsh field research and plant identification. Other technicians should be trained to fulfill responsibilities in the field and adhere to QA/QC procedures. Two field crews should initially work together to collect field data until consistent results are achieved between the two teams, after which the teams should work independently.

8.Preventative Maintenance

Field equipment should be inspected by the crew each day before going out to collect field data. At the field site, equipment testing is advised prior to data collection to ensure that it is working properly. Equipment should be maintained regularly as needed and as recommended by the manufacturer. Laser levels should be calibrated once a month to within manufacturer's specifications (0.5 cm over 100 M). GPS accuracy should be assessed once a month by a check of any units used in the field with a known location.

9.Corrective Actions

Data quality control ensures high quality data however any sites within the same season or period of monitoring which contain data anomalies should be re-assessed. Any sites that

contain anomalous data that cannot be resolved should be flagged and identified in the meta data set.

10. Important Considerations

Samplers new to vegetation assessment will be unsure of what to call live and dead. If any part of the plant is alive, it is considered entirely alive. In plots with *Juncus gerardii* or *Festuca rubra*, plants may have senesced and appear to be dead, but please realize these perennials remain alive and will resprout in fall or next spring. Some care must be taken to ascribe cover to live plants that appear dead and the dead plants that are still rooted.

Rarely, there is an even mix of *S. patens*, *D. spicata* and *J. gerardii* and it can be difficult to tease apart the cover of each. One technique is to pull up 20-30 shoots in a similar area that is outside of the plot and spend a minute separating each species and then looking at the relative abundance of each to inform the percentage cover of each species in the plot.

Some monitoring programs worry that monitors bring biases with estimation of ocular cover and changes in personnel will lead to poor statistical outcomes. To enhance repeatability, some programs (Rocks and Stevens 2018) like to use a simplification of ocular cover which places percentage ranges in classes (e.g., Braun-Blanquet 1965, and others). The use of cover classes is not consistent with this SOP. The centers of these classes, if summed for one plot, rarely equal 100%. Many more samples will be required to find changes in cover classes over time.

To avoid biases between measurers, the point intercept method was developed (Roman et al. 2001). The plot is divided into 25 or 50 sub-plots using strings or sticks and at the intersection of each divider a narrow rod (bayonet) is lowered vertically to the soil and all species that touch the rod are marked on the data sheet, which has one entry for each intersection and species (each plot has a unique datasheet that is a matrix of 25 or 50 intersections and species (expected species are typically pre-loaded to data sheets).

For the point intercept method, the number of 'hits' for each species is multiplied by 4 or 2 (depending on the number of intercepts) to approach values like percent cover. However, these are presence-absence results and often sum to over 100 'percent', so to generate a % cover estimate for each species a plot-specific correction factor needs to be applied. Further, bare is only counted if no live plants are 'hit', so there is inherent bias against this cover class. A complete guide on conversion of point intercept to ocular cover is given in:

<https://nerrsciencecollaborative.org/resource/guide-integrate-plant-cover-two-different-methods-point-intercept-and-ocular-cover>.

If compared to ocular cover results, the point intercept plots consistently underestimate non-live cover due to the inherent bias of monitors

in finding a 'hit'. Part of the non-living cover bias can be avoided through proper training, but bare cover remains a problem.

Although a clear preference for ocular cover eliminates all the above corrections, there will be some monitoring programs who feel the need to deviate. The use of point intercept is only consistent with this SOP if a proper conversion is made to sum the cover to 100% and avoid undercounting non-living cover. The exact methods used must be recorded carefully in the metadata.

11. References

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12. Safety Considerations

- Fieldwork should not be conducted during heavy rain events or unsafe conditions such as electrical storms or high wind events.
- Field personnel should not access sites alone without the instruction of a field manager.
- All persons should carry cell phones or other emergency communication devices while sampling. It is recommended they be waterproof or stored in a waterproof case.
- If there is no safe access to a plot point, the field sampling should either be moved to a location with safe access or not be conducted for that site if a replacement location is unavailable.
- If a boat, canoe, or kayak is used to for access, all persons must wear personal flotation devices while on the water.
- All watercraft used must comply with USCG required safety equipment per 46 U.S.C, Chapter 43. An online Vessel Safety Check interactive program is available from USCG at <http://www.uscgboating.org/safety/vsc.htm>.
- Good judgment should be used in selecting clothes and personal protection items. Common items needed include: extra clothing, sunshade, sunscreen, hats, insect repellent, and waterproof knee boots, chest waders or hip waders for highest anticipated depths. Any staff not dressed appropriately for fieldwork should not

participate in the survey. Proper footwear is a must (e.g., no “flip-flops” for field work).

- Good judgment should be used in walking on marsh surfaces; mosquito ditches will be circumvented, or when deemed possible, crossed with caution.
- Each crew member will carry a personal first aid kit.
- No chemicals should be handled by personnel in the field (decontamination will occur in parking areas or after returning from the field).
- Private property should be respected using the following guidelines.
 - Landowner permission to access property should be sought for all sites.
 - If property is in close proximity to buildings or other heavily used areas, the site should not be assessed without landowner permission.
 - Posted property should not be accessed without permission of the landowner.
 - If we do not receive a response from a landowner to our request for permission to access, proceed with the assessment as long as the site is not posted or in close proximity to buildings or other heavily used areas.
 - If asked to leave private property by the landowner, samplers will discontinue work and leave.

13.0 Data Generation, Management and Storage

13.1 Data to be Generated (somewhat redundant with section 6.3)

Field plot data should be recorded on waterproof data sheets (see attached). Data collected for each data sheet should include site name, date, time, names of crew members collecting data and site name. Data collected for each plot should include transect number, plot number, distance along transect, and RTK-GPS point number. Field data sheets should follow the following naming convention for each day of monitoring: SiteName_Day_Month_Year. Before a transect is disassembled after all plots are monitored, a technician should review visual cover, and water, canopy height, thatch height are recorded. If any issues are found, technicians should re-visit plots with errors and remediate any issues.

13.2 Data Handling, Storage and Backup

At the end of each day, completed data sheets should be immediately scanned and uploaded to the project’s Google Drive folder by a field technician. RTK-GPS data should be downloaded

(Microsoft Excel) and uploaded to the Google Drive folder as well. Scanned datasheets and RTK-GPS datasets should be added to the correct folder according to the following hierarchy:

Root folder:

<<SITE NAME>>

> <<YEAR>>

> FIELD MONITORING DATA

> FieldSheet_SiteName_Year_Month_Day

> RTK_SiteName_Year_Month_Day

After the scanned datasheets are stored in the cloud, the field technician should enter all data from each waterproof plot form and the RTK unit into a Google Drive spreadsheet. Data from the plot sheets and RTK (including the spreadsheet) should be backed up using a project-issued external drive. The external hard drive should be updated at the end of every week.

This results in four levels of redundant plot sheet backup: (1) the waterproof copies stored in binders in the lab; (2) scanned PDFs of datasheets stored in respective folders for the site day, month and year in Google Drive; (3) data entered in the Google spreadsheet stored in the same Google Drive folder; and, (4) scanned PDFs of datasheets and downloaded spreadsheets (Microsoft Excel) to an external hard-drive.

13.3 Project permanent archive

The project should use the Open Science Framework (<http://osf.io>) cloud-based system to store all final project documentation and final datasets. Final datasets include classification outputs, accuracy assessments, maps, etc. This cloud-based system should not be included as a routine data management backup folder. OSF has established a site preservation fund and advertises that the fund is sufficient to ensure the site is up and operational for 50+ years (see <http://help.osf.io/m/faqs/l/726460-faqs>).

14. Appendices

Appendix A: *Guidance on selection of number of plots*

Whereas studies showing changes over time or ascribed to restoration actions have been shown using both univariate (Donnelly and Bertness 2001) and multivariate approaches (Burdick et al. 2019), only one focused on determining the correct number of samples to use for a monitoring program to find vegetation change on salt marsh (James-Pirri et al. 2007). The guidance provided by the diagram is based on randomly selected plots of 1m² within salt marshes (low and high marsh mixed). The generalized guidance from James-Pirri and colleagues suggested 20 plots be gathered from randomized locations (permanent or non-permanent) to show subtle differences that might be expected over two consecutive years.

The protocol here is to use 0.5 or 1.0 m² plots along permanent transects (hence permanent plots) over more than 2 years and only compare plots within marsh zones (low, high and upper edge). These considerations can allow the relaxation of the 20-plot or 20 m² sampling standard indicated by James-Pirri and colleagues in 2007. Although a new power analysis based on permanent plots is needed, we believe the total sampling area can be reduced to 10 m² and the number of plots collected in a specific marsh zone or for a specific restoration action can be reduced accordingly.

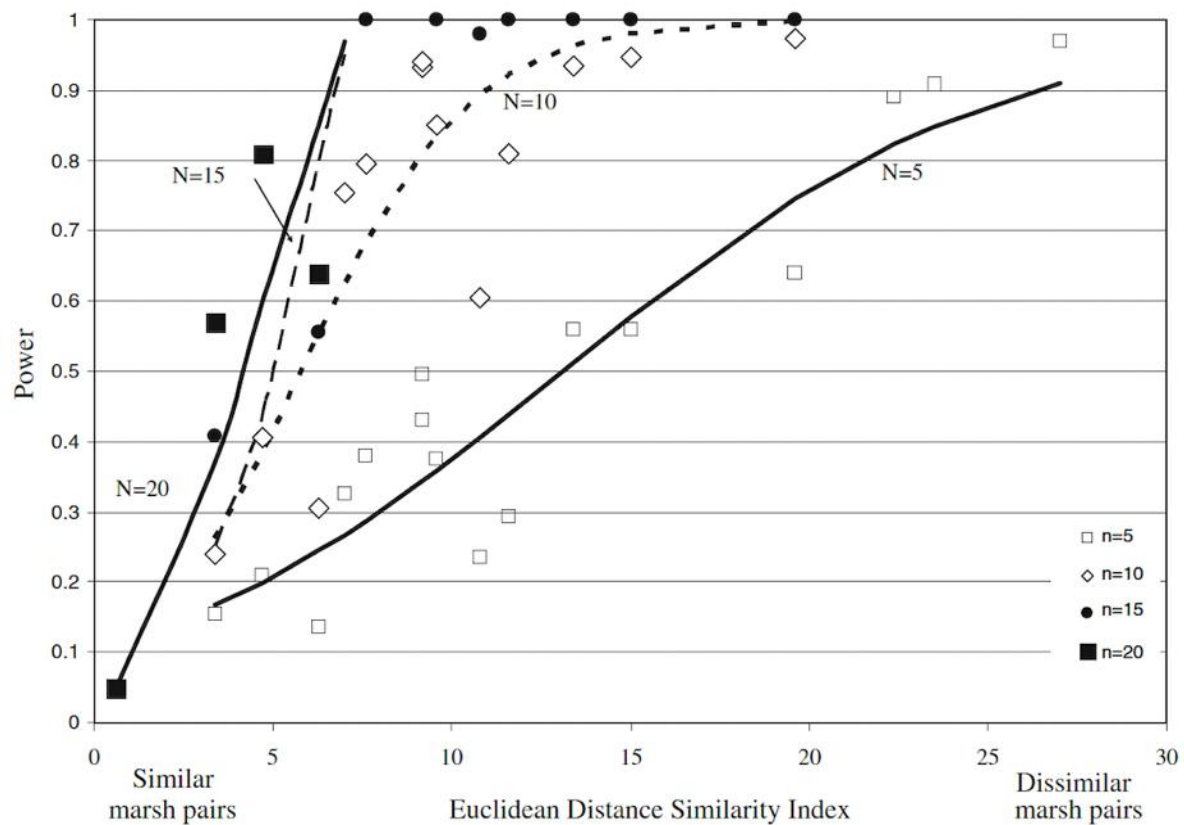


Figure A-1. From James-Pirri et al. 2007, power curves for varying sample sizes along a gradient of community similarity.

Appendix C: Field and spreadsheet data sheets with examples

Salt Marsh Monitoring – Vegetation Survey Data Sheet

Site, Town, State _____ Date: _____ Field Monitors: _____

Quadrat Size: _____ Other Info: _____

Transect	Plot	Distance	Habitat (UE=Upper Edge)				%	Species	%	Species	%	Species	%	Plant Height (cm)	
			High	Pool											
			High	Pool		Bare									
			Trans.	Panne		Dead									
			Low	UE		Wrack									
			High	Pool		Bare									
			Trans.	Panne		Dead									
			Low	UE		Wrack									
			High	Pool		Bare									
			Trans.	Panne		Dead									
			Low	UE		Wrack									
			High	Pool		Bare									
			Trans.	Panne		Dead									
			Low	UE		Wrack									
			High	Pool		Bare									
			Trans.	Panne		Dead									
			Low	UE		Wrack									
			High	Pool		Bare									
			Trans.	Panne		Dead									
			Low	UE		Wrack									
			High	Pool		Bare									
			Trans.	Panne		Dead									
			Low	UE		Wrack									
			High	Pool		Bare									
			Trans.	Panne		Dead									
			Low	UE		Wrack									

Salt Marsh Monitoring – Vegetation Survey Data Sheet

Site, Town, State Crane Estate, Ipswich, MA Date: 7/30/2024 Field Monitors: John Doe, Jane Buck

Quadrat Size: 0.5m² Other Info: _____

Transect	Plot	Distance	Habitat (UE=Upper Edge)				%	Species	%	Species	%	Species	%	Canopy Height (cm)	Thatch Height (cm)	Notes
IP-1	1	0	High		Pool		Bare	12	Spalt	75				73	none	
			Trans.		Panne		Dead	8	Linac	5						
			Low	x	UE		Water	0	Wrack	0						
IP-1	2	12	High	x	Pool		Bare	3	Spalt	12	Sadep			12	4	
			Trans.		Panne		Dead	12	Sppat	45	Dispi	25				
			Low		UE		Water	0	Wrack	3						
IP-1	3	24	High		Pool		Bare	34	Spalt	45				29	none	
			Trans.		Panne	x	Dead	12	Sumar	6	Sadep	3				
			Low		UE		Water	30	Wrack							
IP-1	4	36	High		Pool	x	Bare	100						-	none	
			Trans.		Panne		Dead									
			Low		UE		Water	100	Wrack							
IP-1	5	48	High	x	Pool		Bare	12	Sppat	60				16	5	
			Trans.		Panne		Dead	8	Juger	8						
			Low		UE		Water	0	Wrack	12						
IP-1	6	53.5	High		Pool		Bare	8	Sppec	6	Dispi	12		54	none	
			Trans.		Panne		Dead	16	Pavir	12	Ivfru	15				
			Low		UE	x	Water	0	Wrack	25	Torad	6				
			High		Pool		Bare									
			Trans.		Panne		Dead									
			Low		UE		Water		Wrack							
			High		Pool		Bare									
			Trans.		Panne		Dead									
			Low		UE		Water		Wrack							
			High		Pool		Bare									
			Trans.		Panne		Dead									
			Low		UE		Water		Wrack							

Spreadsheet of data.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
Date	Site	Transect	Distance (m)	Transect Type	Impact Treatment	Habitat	Plot location notes	TOTAL	Bare Mud	Dead	Water	Algal Mat	Wrack
9/23/21	South Runnel	T1	0	RUN	No	HM	East Post	100	4	5			
9/23/21	South Runnel	T1	1.7	RUN	Yes	PE	Transition	100	18	2			

P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM	
<i>Spartina alterniflora</i>	<i>Spartina patens</i>	<i>Distichlis spicata</i>	<i>Juncus gerardii</i>	<i>Salicornia depressa</i>	<i>Vaucheria</i>	<i>Agalinas maritima</i>	<i>Atriplex patula</i>	<i>Festuca rubra</i>	<i>Glaux maritima</i>	<i>Iva frutescens</i>	<i>Limonium nashii</i>	<i>Panicum virgatum</i>	<i>Plantago maritima</i>	<i>Polygonum ramosissimum</i>	<i>Puccinellia</i> sp.	<i>Ruppia maritima</i>	<i>Schoenoplectus maritimus</i>	<i>Schoenoplectus robustus</i>	<i>Solidago sempervirens</i>	<i>Spartina pectinata</i>	<i>Spergularia marina</i>	<i>Suaeda linearis</i>	<i>Suaeda maritima</i>	<i>Synproctium tenuifolium</i>	<i>Triglochin maritimum</i>	<i>Agarophyton vermiculophyllum</i>	<i>Algae</i>	<i>Ascomyllum nodosum</i>	<i>Filamentous algae</i>	<i>Fucus spiralis</i>	<i>Fucus vesiculosus</i>	<i>Ulva</i> Spp.	<i>Agrostis stolonifera</i>	<i>Agropyron repens</i>	<i>Agrostis alba</i>	<i>Ambrosia</i> sp.	<i>Amelanchier canadensis</i>	<i>Ammophila breviligulata</i>	<i>Annaphalis</i> sp.	<i>Artemisia vulgaris</i>	<i>Aster</i> sp.	<i>Aster subulatas</i>	<i>Calamagrostis canadensis</i>	<i>Calystegia sepium</i>	<i>Chenopodium album</i>	<i>Chenopodium rubrum</i>	<i>Clover</i>	<i>Digitaria grass</i> sp.	<i>Echinola crus-gali</i>	
	70	20					1																																											
20	45	15																																																

BN	BO	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD	CE	CF	CG	CH	CI	CJ	CK	CL	CM	CN	CO	CP	CQ	CR	CS	CT	CU	CV	CW
<i>Elymus spp</i>	<i>Elymus virginicus</i>	<i>Juncus balticus</i>	<i>Lythrum salicaria</i>	<i>Myrica gale</i>	<i>Myrica pensylvanica</i>	<i>Phalaris arundinacea</i>	<i>Phragmites americanus</i>	<i>Phragmites australis</i>	<i>Polygonum sp.</i>	<i>Rosa multiflora</i>	<i>Rosa rugosa</i>	<i>Rubus sp.</i>	<i>Rumex crispus</i>	<i>Schoenoplectus olneyii</i>	<i>Schoenoplectus pungens</i>	<i>Schoenoplectus validus</i>	<i>Schoenoplectus spp.</i>	<i>Solidago rugosa</i>	<i>Solidago sp.</i>	<i>Steria sp.</i>	<i>Teucrium canadense</i>	<i>Toxicodendron radicans</i>	<i>Typha angustifolia</i>	<i>Typha latifolia</i>	<i>Vetch</i>	<i>Unknown 1</i>	<i>Unknown 2</i>	<i>Unknown 3</i>	<i>Canopy Height_4/5 cm</i>	<i>Percent Live</i>	<i>Percent Not Live</i>	<i>Percent Halo</i>	<i>Percent Algae Total</i>	<i>Vascular Richness Total</i>	<i>Richness HALO</i>
																													18	91	9	91	0	3	3
																													28	80	20	80	0	4	4

List of marsh plant species and five-letter codes.

Common Name	Latin Name	Habitat	Code	Common Name	Latin Name	Habitat	Code
Seaside Arrowgrass	<i>Triglochin maritima</i>	High Marsh, Pannes	TRMAR	Red Fescue	<i>Festuca rubra</i>	High Marsh	FERUB
American Germander	<i>Teucrium canadensis</i>	Marsh Border	TECAN	Reed Canary Grass	<i>Phalaris arundinacea</i>	High Marsh	PHARU
Baltic Rush	<i>Juncus balticus</i>	Brackish Areas	JUBAL	Rugosa Rose	<i>Rosa rugosa</i>	Marsh Border	RORAG
Black Grass	<i>Juncus gerardii</i>	High Marsh	JUGER	Salt Marsh Aster	<i>Symphotrichium tenuifolius</i>	High Marsh	SYTEN
Broad-Leaf Cattail	<i>Typha latifolia</i>	Brackish Areas	TYLAT	Salt Marsh Bulrush	<i>Bolboschoenus maritimus</i>	Brackish Areas	BOMAR
Bushy Knotweed	<i>Polygonum ramosissimum</i>	Marsh Border	PORUM	Salt Marsh Bulrush	<i>Bolboschoenus robustus</i>	Brackish Areas	BOROB
Chairmaker's Rush	<i>Schoenoplectus americanus</i>	Brackish Areas	SCAME	Salt Marsh Gerardia	<i>Agalinas maritima</i>	High Marsh	AGMAR
Common Glasswort	<i>Salicornia depressa</i>	Pool/Panne	SADEP	Salt Meadow Grass	<i>Spartina patens</i>	High Marsh	SPPAT
Common Reed	<i>Phragmites australis</i>	Marsh Border	PHAUS	Saltwater Cordgrass	<i>Spartina alterniflora</i>	Low Marsh	SPALT
Creeping Bent Grass	<i>Agrostis stolonifera</i>	High Marsh	AGSTO	Sea Blite	<i>Sueada linearis</i>	Pool/Panne	SULIN
Groundsel Tree	<i>Baccharis halimifolia</i>	Marsh Border	BAHAL	Sea Lavender	<i>Limonium carolinianum</i>	High Marsh	LICAR
Hedge Bindweed	<i>Calystegia sepium</i>	Marsh Border	CASEP	Sea Milkwort	<i>Glaux maritima</i>	High Marsh/Panne	GLMAR
Jewelweed	<i>Impatiens capensis</i>	Brackish Areas	IMCAP	Seashore Alkali Grass	<i>Puccinellia maritima</i>	High Marsh	PUMAR
Marsh Elder	<i>Iva frutescens</i>	Marsh Border	IVFRU	Seaside Arrow Grass	<i>Triglochin maritima</i>	High Marsh	TRMAR
Marsh Orach	<i>Atriplex patula</i>	High Marsh	ATPAT	Seaside Goldenrod	<i>Solidago sempervirens</i>	High Marsh	SOSEM
Marsh Rush	<i>Juncus effusus</i>	Brackish Areas	JUEFF	Seaside Plantain	<i>Plantago maritima</i>	Pool/Panne	PLMAR
Meadowsweet	<i>Spiraea latifolia</i>	Marsh Border	SPLAT	Silverweed	<i>Argentina anserina</i>	High Marsh	ARANS
Narrow-Leaf Cattail	<i>Typha angustifolia</i>	Brackish Areas	TYANG	Spike Grass	<i>Distichlis spicata</i>	High Marsh	DISPI
Northern Bayberry	<i>Myrica pennsylvanica</i>	Marsh Border	MYPEN	Sweet Gale	<i>Myrica gale</i>	Marsh Border	MYGAL
Poison Ivy	<i>Toxicodendron radicans</i>	Marsh Border	TORAD	Switchgrass	<i>Panicum virgatum</i>	Marsh Border	PAVIR
Prairie Cordgrass	<i>Spartina pectinata</i>	Brackish Areas	SPPEC	Unknown Species	<i>Unknown Species</i>		UNK
Purple Loosestrife	<i>Lythrum salicaria</i>	Brackish Areas	LYSAL	Widgeon Grass	<i>Rupia maritima</i>	Pool	RUMAR